Supplementary Material: Coordination-Collapse Phase Boundary in Policy-Gradient Cooperative MARL

Anonymous submission

This document contains the apparatus supporting the main paper: full experimental setup and hyperparameters (§1), the complete loss-aversion sweep and baseline reconciliation (§2), the full mean-field derivation and all three theorems (§3), the cross-environment and cross-algorithm detail (§4), both boundary-condition ablations (§5), the pre-registered signal-to-noise probe (§6), the exploration-noise ablation (§7), threshold robustness (§8), notation (§9), and a claim–evidence ledger (§10). Section, equation, and figure references to the main paper are by name. All designs were pre-registered before execution (commit hash withheld for anonymity, released on acceptance).

1 Experimental Setup in Full

Primary substrate. `NetworkManagement-v1` from or-gym (Hubbs et al. 2020): a six-node multi-echelon inventory network with Poisson demand (μ=20) on the retail edge, 30-period episodes, uniform lead times. Node 1 (retail-facing) follows a fixed demand-pull policy, stabilizing the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterized via a base-stock wrapper mapping each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \text{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work, the Beer Distribution Game Dec-POMDP of Oroojlooyjadid et al. (2022), the multi-echelon A3C results of Gijsbrechts et al. (2022), and single-echelon MARL (Leluc et al. 2023), to a six-node network with per-node agents.

Algorithm and evaluation. Each agent runs PPO (Schulman et al. 2017) with a shared MLP policy (`n_steps`=128, `batch_size`=256), implemented with Stable-Baselines3 (Raffin et al. 2021) and PettingZoo/SuperSuit vectorization (Terry et al. 2021; Terry, Black, and Hari 2020). Evaluation is deterministic over 10 episodes with 20 seeds per condition.

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Table 1: Primary experiment summary (same environment and algorithm; α=β=1 is the linear no-PT condition).

Exp	Purpose	λ	α, β	Seeds	Runs
A @150k	Struct. boundary	1	1,1	20	220
A @300k	Struct. boundary	1	1,1	20	140
C	Basin control	1	1,1	12	144
N	Null baseline	1	1,1	20	180
B-Red.	Behav. decomp.	1,3,7	0.88	20	300
λ-sweep	Loss-av. law	2,4,5,6	0.88	20	400
T (600k)	Metastability	1	0.88	20	140

Table 2: The recurring w_{crit} values and the single role each plays.

w_{crit}	Condition	Role
0.666	α=β=1, 300k	canonical null
0.685	α=β=1, 300k (Exp N)	null, reproduced
0.822	α=β=1, 150k	null at shorter horizon
0.586	α=β=0.88, λ=1	curvature result

Experiment inventory. Table 1 summarizes the seven primary experiments; total run count 1,524.

Basin control (rules out initialization). Exp C rules out initialization as a collapse driver. Within-w correlations between S_{init} and converged S are -0.11 (w=0.5), -0.38 (w=0.85), and -0.36 (w=1.0), all near zero and inconsistent with initialization dominating; the collapsed/survived assignment is insensitive to where agents start.

2 Full Behavioral Sweep and Baseline Reconciliation

The recurring w_{crit} values. There is one structural null; the other recurring numbers are a horizon measurement of it and a curvature *result*, not independent baselines (Table 2).

Full loss-aversion sweep. Table 3 gives the complete λ=1…7 sweep with Firth penalized fits. The increase is monotone up to estimation noise at the

Table 3: Full loss-aversion sweep (α=β=0.88, 300k, 20 seeds/(w, λ) cell, n=100 runs/λ). w_{crit} with bootstrap 95% CIs; slope/p from Firth penalized logistic regression.

λ	w_{crit}	95% CI	Firth b_1	p
1	0.586	[0.539, 0.624]	-14.7	1.2×10^{-6}
2	0.715	[0.687, 0.745]	-27.2	4.4×10^{-6}
3	0.774	[0.751, 0.798]	-29.7	5.1×10^{-6}
4	0.806	[0.766, 0.841]	-36.8	5.5×10^{-6}
5	0.825	[0.806, 0.844]	-53.1	4.7×10^{-4}
6	0.835	[0.817, 0.850]	-24.7	2.5×10^{-6}
7	0.852	[0.841, 0.864]	-43.1	1.5×10^{-3}

high-λ plateau: w_{crit} rises strictly from λ=1 to λ=5 then saturates near 0.84–0.85 (λ=5, 6 statistically indistinguishable). At high loss aversion the transition is very steep and falls between adjacent grid points (collapse fractions $1.00, 1.00, 1.00, 0.55, 0.00$ at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$ for λ=7), so the boundary remains identifiable (the w=0.85 cell is interior, 11/20 collapsed) but the unpenalized MLE slope diverges. Firth penalized logistic regression is stable here; all seven slopes are significant ($p < 10^{-3}$) and the Firth w_{crit} matches the bootstrap to within 0.019, confirming the large unpenalized slopes at λ=5, 7 were a fitting artifact, not separation. The high-λ plateau is a behavioral saturation, not a w=1 ceiling artifact: at λ=7 coordination is fully restored (collapse fraction 0.00) already at w=0.95, consistent with the bounded affine law in which $w_{\text{crit}}(\lambda) \to 1$ only as $\lambda \to \infty$. The $\lambda \geq 5$ values in Table 3 use a finer $w \in \{0.80, 0.90\}$ grid added specifically to resolve the steep transition; this places two interior points across each high-λ boundary and halves the CI widths ($0.038, 0.033, 0.024$ at λ=5, 6, 7) relative to the single w=0.85 cell, giving the clean monotone $0.825 \to 0.835 \to 0.852$.

Metastability under matched curvature. The 600k row of the main-paper heatmap uses α=β=0.88, which shifts w_{crit} left by 0.100, so those collapse fractions are partly curvature-deflated. Under matched α=β=1 (Exp 0b, 20 seeds, 600k), $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$, versus 0.80 and 0.40 at 300k: the horizon slide holds with curvature held constant; only the figure's 600k magnitude is curvature-inflated.

The metastable slide generalizes. The horizon dependence is not a supply-chain artifact. We ran dedicated horizon sweeps under centralized-critic MAPPO on both transfer environments. On MPE Cooperative Navigation, w_{crit} falls $0.82 \to 0.68$ as the horizon grows 2M $\to$ 8M env steps (slide +0.14); on Level-Based Foraging it falls $0.33 \to 0.20$ over 1M $\to$ 4M (slide +0.13), each with non-overlapping bootstrap intervals at the endpoints. Both match the supply-chain direction (boundary slides left with more training), so the metastability result (like the loss-aversion direction) holds across environments and algorithms, not just on the primary substrate.

Table 4: Matched risk-neutral horizons (α=β=1, λ=1). w_{crit} with bootstrap 95% CIs; the four points fit $w_{\text{crit}} = 2.44 - 0.138 \ln T$ at RMSE 0.027.

T (env steps)	w_{crit}	95% CI
150k	0.822	[0.785, 0.858]
300k	0.666	[0.635, 0.694]
450k	0.622	[0.603, 0.641]
900k	0.573	[0.547, 0.598]

The four-horizon log-linear fit. On the supply chain we measured w_{crit} at four matched risk-neutral horizons (Table 4). The four points fall on a log-linear law $w_{\text{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027), the Kramers escape-time prediction that w_{crit} falls as $-\ln T$ toward the deterministic floor $w^0_{\text{crit}} = c/(c+g)$. The two longer horizons (450k, 900k) have bootstrap CIs of width $\leq$0.04, so the law is a fitted four-point relationship, not a two-point extrapolation.

The derived affine law (7) and its out-of-sample fit (RMSE 0.014 across all seven points) are shown and discussed in the main paper; the derivation of its form is given in §3 below.

3 Mean-Field Mechanism in Full

This section develops the mechanism the main paper summarizes. It is a mechanistic account of why the boundary exists and how it moves, not a closed-form derivation of w_{crit}. We are explicit about what is derived, what is calibrated, and what is not claimed (Table 5). As noted in the main paper, the deep-MARL contribution is the learning-dynamics analogue of the established EGT result (Mengesha, Sun, and Roy 2025): here the driver is the signal-to-noise ratio of the policy gradient under exploration, not selection pressure over fixed strategies.

3.1 Gradient flow and its noise

Each agent i runs PPO on a Gaussian policy over its normalized base-stock action $a_i \in [0, 1]$ with mean μ_i and exploration scale σ. The shaped reward is the centralization blend through the CPT value function v,

$$r_i = v\big((1-w)\,\ell_i(a_i, a_{-i}) + w\,G(a)\big), \quad G(a) = \textstyle\sum_j \ell_j, \tag{1}$$

with ℓ_i node i's local profit and G system profit. For a Gaussian policy the PPO mean update is a stochastic estimate of the return gradient, so the mean follows a mean-field Langevin (Fokker–Planck) equation

$$\mathrm{d}\mu_i = \underbrace{\eta\,\partial_{\mu_i}\mathbb{E}[(1-w)\ell_i + wG]}_{\text{drift } D(w)}\,\mathrm{d}t + \sqrt{2\,\Theta(w)}\,\mathrm{d}W. \tag{2}$$

Because each partner's local profit is blended into the global term, partner exploration σ enters agent i's gradient estimate through the global channel, so the diffusion carries a centralization-weighted cross-agent piece

$$\Theta(w) \propto \tfrac{\sigma^2}{B}\big[(1-w)^2 s_\ell^2 + w^2(N-1)\kappa^2\big], \tag{3}$$

with batch size B, local sensitivity s_ℓ, cross-agent sensitivity κ.

3.2 The boundary is a signal-to-noise crossing

A direct finite-difference probe of ℓ_i and G along the base-stock axis confirms the asymmetry the model needs: an upstream node's inventory is a *local cost* (holding and procurement reduce its profit) but a *global benefit* (it averts downstream stockouts). Writing the linearized marginals $\partial_a \mathbb{E}\ell \to -c$, $\partial_a \mathbb{E}G \to +g$ $(c, g > 0)$, the cooperative drift is

$$D(w) = \eta\big[wg - (1-w)c\big]. \tag{4}$$

Escape from the lazy basin requires $\mathrm{SNR}(w) = D(w)/\sqrt{\Theta(w)} \gtrsim \kappa_c$ for an $\mathcal{O}(1)$ critical κ_c set by the training budget; the crossing defines w_{crit}. In the low-noise limit $\Theta \to 0$ this reduces to $D(w) = 0$, i.e. $w^0_{\mathrm{crit}} = c/(c+g)$, the asymptotic $(T \to \infty)$ boundary; finite exploration noise raises the operative boundary above it. The coefficients c, g, σ, κ_c are not derived in closed form from the multi-echelon cost matrix; the model is calibrated and its content is the functional predictions, not the magnitude.

3.3 Sharpness and metastability

Escape is a Kramers first-passage problem, $\tau \sim \tau_0 \exp(\Delta U/\Theta)$, exponentially sensitive to SNR, so collapse probability switches abruptly across w_{crit} (matching the steep logistic fits). Collapse at finite horizon means $\tau > T$; as T grows a smaller SNR (hence smaller w) suffices, so $w_{\mathrm{crit}}(T)$ decreases monotonically toward w^0_{crit}, the observed $0.822 \to 0.666$ slide, following the $\propto -\ln T$ Kramers trend.

3.4 Formal account: predicted versus calibrated

Let $x \in [0,1]$ be a macroscopic order parameter (fraction of agents coordinating) evolving as

$$\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t, \tag{5}$$

with F the mean-field drift (so $\partial_w F > 0$ and $\partial_\lambda F < 0$ follow from the payoff structure: w weights the cooperation gradient $+g$, λ scales only the loss domain) and $\Sigma > 0$ the gradient-noise variance. x is *latent*: consistent with our pre-registered probe (§6) it is not recoverable from policy statistics, so the claims concern its law, not a measured trajectory.

Theorem 1 (Mean-field criticality, $N \to \infty$)**.** *If $\Sigma(x, N) = D(x)/N$, then as $N \to \infty$ the dynamics become deterministic, $\dot{x} = F(x, w, \lambda)$, and the coordinating basin disappears via a saddle-node bifurcation. The mean-field threshold $w^{\mathrm{MF}}_{\mathrm{crit}}$ is the unique solution of $F(x^*, w, \lambda) = 0$, $\partial_x F(x^*, w, \lambda) = 0$, a closed-form function of the payoff and prospect-theory parameters.*

Proof. With $\Sigma = D(x)/N \to 0$ the noise in (5) vanishes and the flow is $\dot{x} = F$. A coordinating stable node and an unstable saddle bound the basin; the basin vanishes when they collide, i.e. $F = 0$ and $\partial_x F = 0$ coincide (saddle-node). As F is closed-form in the payoff and CPT functions, so is its root $w^{\mathrm{MF}}_{\mathrm{crit}}$. □

Theorem 2 (Calibration gap, finite N and horizon T)**.** *For finite N with correlated CTDE updates, $\Sigma(x, N) > 0$, and collapse is a noise-induced escape rather than a bifurcation. The empirical boundary is the metastability root $\tau(w, \lambda) = T$ of the Kramers mean first-passage time*

$$\tau \asymp \exp(\Delta U), \quad \Delta U = \int_{x^*_{\mathrm{coord}}}^{x^*_{\mathrm{saddle}}} \frac{-2F(z, w, \lambda)}{\Sigma(z, N)}\,\mathrm{d}z, \tag{6}$$

and $w^{\mathrm{Emp}}_{\mathrm{crit}} \neq w^{\mathrm{MF}}_{\mathrm{crit}}$ in general; the offset cannot be isolated without measuring Σ.

Proof. The stationary density of (5) is $P_{ss}(x) \propto \exp[-U_{\mathrm{eff}}(x)]$ with $U_{\mathrm{eff}}(x) = -\int_0^x 2F/\Sigma\,\mathrm{d}z$, giving the barrier ΔU of (6); Kramers' rate yields $\tau \asymp e^{\Delta U}$. At finite T, collapse occurs when $\tau \le T$, so the boundary solves $\tau(w, \lambda) = T$. Since Σ is the intractable covariance of PPO updates, ΔU and hence $w^{\mathrm{Emp}}_{\mathrm{crit}}$ require calibration. The two notions coincide as limits: finite (N, T) escapes before the Theorem 1 bifurcation, and $w^{\mathrm{Emp}}_{\mathrm{crit}}(T) \downarrow w^{\mathrm{MF}}_{\mathrm{crit}}$ as $T \to \infty$. □

Proposition 1 (Sign propagation of the loss-aversion shift)**.** *Assume $\Sigma > 0$ and the payoff premise $\partial_\lambda F(x, w, \lambda) < 0$ for all $x \in [x^*_{\mathrm{saddle}}, x^*_{\mathrm{coord}}]$ (verified by finite-difference probe; §3). Then $\partial_\lambda w^{\mathrm{Emp}}_{\mathrm{crit}} > 0$.*

Proof. Write (6) as a forward integral $\Delta U = \int_{x^*_{\mathrm{saddle}}}^{x^*_{\mathrm{coord}}} 2F/\Sigma\,\mathrm{d}z$ (the two minus signs cancel). Differentiating under the integral, fixed points contribute nothing to first order ($F = 0$ there), so $\partial_\lambda \Delta U = \int 2\,\partial_\lambda F/\Sigma\,\mathrm{d}z < 0$ pointwise because $\Sigma > 0$ and $\partial_\lambda F < 0$ on the whole interval, independent of the sign of F. A lower barrier shortens escape time, $\partial_\lambda \tau < 0$; symmetrically $\partial_w F > 0$ gives $\partial_w \tau > 0$. The implicit function theorem on $\tau(w, \lambda) - T = 0$ gives $\mathrm{d}w^{\mathrm{Emp}}_{\mathrm{crit}}/\mathrm{d}\lambda = -\partial_\lambda \tau/\partial_w \tau > 0$. The magnitude depends on Σ (Theorem 2), but the direction is guaranteed by the drift alone. The identical argument with ∂F signs reversed for concave–convex curvature gives the opposite shift. □

Table 5: What the mean-field mechanism claims (extended version of the main-paper ledger, adding the exploration-noise row and the MPE-null detail).

Derived (no fit)	opposite signs of curvature vs. loss aversion; monotone $w_{\text{crit}}(\lambda)$; affine $\phi(\lambda)$ with $\rho \in [0,1]$; phase-transition sharpness; w_{crit} decreasing with T; w_{crit} increasing with exploration noise.
Empirically confirmed	opposite signs and affine form; horizon slide (also on MPE, LBF); exploration-noise dependence (supply chain and LBF; predicted null on MPE).
Calibrated	g/c; loss-domain share ρ; noise scale setting absolute w_{crit}.
Not claimed	closed-form w_{crit} from the cost matrix; numerical magnitude on any axis; the mild concavity beyond leading order.

3.5 Opposite signs and the derived affine law

Concave–convex curvature ($\alpha, \beta < 1$) compresses both gains and losses, lowering the variance entering Θ, raising the SNR, and *lowering* w_{crit}; loss aversion ($\lambda > 1$) magnifies the downside, inflating Θ, lowering the SNR, and *raising* w_{crit}. The signs are fixed before any data. The functional form follows by splitting the marginal mass into gain/loss domains; only the loss domain scales with λ, giving

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \quad \phi(\lambda) = 1 + \rho(\lambda - 1), \qquad (7)$$

with $\rho \in [0,1]$ bounded by the theory. Fixing g/c from the λ=1 baseline and $\rho = 0.61$ from three calibration points ($\lambda \in \{1,3,7\}$), held fixed and tested out of sample against $\lambda \in \{2,4,5,6\}$, gives $w_{\text{crit}}^{\text{pred}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\}$ versus $w_{\text{crit}}^{\text{meas}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$, RMSE 0.014 (plotted in the main paper). The one systematic residual is mild concave overshoot at the plateau, consistent with agents endogenously reducing loss exposure as λ grows.

4 Generalization Detail

4.1 Cross-environment: MPE Cooperative Navigation

Three agents cover three landmarks in a continuous 2-D arena (discrete actions, 50-step episodes). The locally rational action under the herding incentive is to stay near teammates, collectively disastrous, as agents cluster and no landmark is covered. We set `local_ratio`=0 so the raw reward is the pure global team score; the herding term is defined by our wrapper. Collapse is per-step landmark occupancy (fraction of landmarks within a fixed radius of some agent), a behavioral metric invariant to reward scale and so unaffected by the CPT transform. Unlike the supply chain (per-agent incentives), MPE's coordination reward is shared-global, which under independent learners confounds behavioral collapse with credit-assignment failure; we therefore train MPE with centralized-critic MAPPO (Yu et al. 2022; Lowe et al. 2017). We set the herding scale so the null boundary is interior (w_{crit}=0.74) and ran the full λ-sweep (1,760 runs, 20 seeds/cell; Figure 1). The loss-aversion effect replicates, w_{crit} rises 0.78 → 0.89 over λ=1 → 7, rising in envelope but non-monotone at intermediate λ, excess +0.158 [0.108, 0.212] at λ=7 and +0.101 [0.055, 0.150] at λ=3 (matching the supply chain's +0.089). The curvature effect does *not* replicate (+0.042 [−0.009, 0.092], opposite in sign), and the affine form fits MPE only to RMSE 0.05.

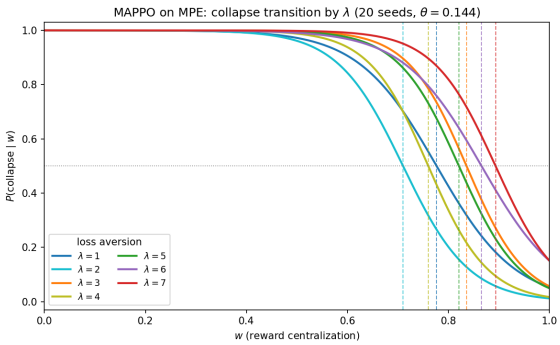


Figure 1: MAPPO on MPE Cooperative Navigation: fitted $P(\text{collapse} \mid w)$ for λ=1,…,7 (α=β=0.88, centralized critic, collapse = landmark occupancy below θ=0.144). The family shifts rightward with λ, the same loss-aversion signature, under a different environment and algorithm.

4.2 A third environment: Level-Based Foraging

LBF (Christianos, Schäfer, and Albrecht 2020) has a public-goods incentive: agents pay an effort cost to load food cooperatively, so the locally rational action is to free-ride. Natively discrete, unrelated to logistics or navigation, trained under centralized-critic MAPPO (matching MPE so only the environment varies), full λ-sweep, same collapse-midpoint criterion (1,600 runs). The risk-neutral null is low and tight ($w_{\text{crit}} \approx 0.08$); curvature moves it *right* by +0.18 (opposite to the supply chain). Loss aversion drives w_{crit} from 0.26 at λ=1 to 0.83 at λ=4, monotone with non-overlapping endpoint CIs (the pre-registered criterion), robust to the collapse threshold. For $\lambda \geq 5$ the boundary saturates against the centralization ceiling: the logistic w_{crit} extrapolates to $w \geq 1$ (outside the measured $w \in [0, 0.95]$ grid), so these points are *not* interior estimates and we label them exploratory in the precise sense that their

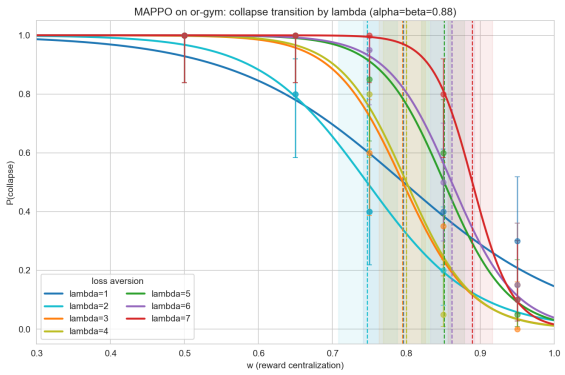


Figure 2: Centralized-critic MAPPO on the supply chain: $P(\text{collapse} \mid w)$ for λ=1, . . . , 7 (700 runs). As under PPO, the family shifts rightward with loss aversion, confirming the effect is not an artifact of independent PPO.

w_{crit} lies beyond the range any run samples, their location is bounded below by 0.95 but not upper-bounded by the data. An added intermediate level w=0.90 tightens the mid-λ estimates and confirms the saturation, but does not bring the $\lambda \geq 5$ crossings inside the grid. The supply chain's affine law does not transfer: with g/c fixed from λ=1, the best-fit slope saturates at its bound and misfits LBF by RMSE 0.15, converting the form's substrate-specificity from assertion to measured result.

4.3 Cross-algorithm: MAPPO on the supply chain

To isolate the algorithm from the environment, we rerun the full sweep on the *identical* substrate, grid, seeds, label, and hyperparameters, changing only the learner from independent PPO to centralized-critic MAPPO (700 runs; Figure 2). w_{crit} rises 0.796 [0.742, 0.860] at λ=1 to 0.889 [0.856, 0.917] at λ=7, monotone from λ=2 (0.747, 0.795, 0.800, 0.851, 0.861, 0.889), endpoint CIs non-overlapping. The direction generalizes while the absolute level does not: MAPPO sits above PPO at low λ (most visibly λ=1, 0.796 vs 0.586, where MAPPO's transition is shallow and the estimate noisier), converging by $\lambda \geq 4$ ($|\Delta w_{\text{crit}}| \leq 0.04$).

5 Two Boundary-Condition Ablations

5.1 Action representation

The mechanism predicts that destroying the continuous policy gradient removes the boundary. Value-based learners need discrete actions, so we discretized the base-stock control into K=11 bins and ran IQL and QMIX on the identical reward pipeline, w-grid, and label; both produced a flat, w-independent collapse rate with no fittable w_{crit}. To isolate the cause, a single-variable control (PPO on the same K=11 discrete shell) also goes flat (Table 6). Discretization *alone* erases the boundary, so the value-based nulls need no appeal to the learning paradigm. This is not a claim that discreteness per se removes the boundary: MPE is natively discrete and retains it. Coarse bins snap the policy to a stable interior level and never visit the near-threshold (s, S) region where collapse occurs, exactly the continuous-gradient prediction. The effect is a coarse-discretization boundary condition, substrate-specific like the curvature sign, and a confirmed prediction, not a negative result.

Table 6: Action-representation ablation. The control (last row) changes only the action space while holding the learner at PPO, isolating coarse discretization, not the value-based paradigm, as what erases the boundary.

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes
QMIX (centralized value)	discrete K=11	no
IQL (independent value)	discrete K=11	no
PPO (control)	discrete K=11	**no**

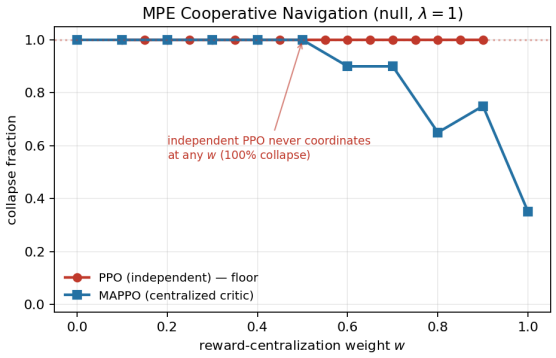


Figure 3: Credit-assignment ablation on MPE (null, λ=1). Independent PPO collapses at every w (a coordination floor), while centralized-critic MAPPO crosses the boundary. The behavioral boundary is exposed only once credit assignment is handled.

5.2 Credit assignment

The boundary is measurable only with a learner that can coordinate on a shared-reward task. On MPE, independent PPO never reaches the coordinated regime at *any* w (collapse fraction 1.0 throughout, including maximal centralization; Figure 3), so behavioral collapse cannot be distinguished from inability to assign credit. Centralized-critic MAPPO recovers coordination (collapse fraction falls to 0.35, occupancy crosses the threshold near $w \approx 0.6$). The same floor appears on LBF. The boundary is therefore measurable only with a centralized critic on shared-reward tasks, why we match one to such environments: a necessity, not a convenience.

6 Pre-registered Signal-to-Noise Probe

The mean-field account calibrates the magnitude of w_{crit} rather than predicting it, because the drift and diffusion coefficients are properties of the stochastic learning dynamics. We pre-registered a direct measurement to upgrade magnitude from calibrated to predicted, fixing its analysis before running. **Observable:** during PPO training we log the mean base-stock action μ once per update and, over a fixed early window, estimate drift $D(w)$ (mean increment) and diffusion $\Theta(w)$ (increment variance/2), forming $\text{SNR}(w) = D(w)/\sqrt{2\Theta(w)}$. **Decision rule:** sweep $w \in \{0.50, \ldots, 0.90\}$ at the structural condition, 20 seeds/w, 150k steps; the mechanism is corroborated iff the measured crossing's bootstrap 95% CI contains the matched-horizon boundary $w_{\text{crit}} = 0.822$. **Outcome (reported in full):** the collapse-fraction of this independent sweep crosses 0.5 at $w = 0.82$, reproducing the matched-horizon boundary almost exactly, a positive robustness check on the boundary. The *drift-sign* observable, however, was *inconclusive*: the restoring drift in the contested band is weakly positive across all w with no genuine zero-crossing, so it does not localize w_{crit}. The reason is basin geometry, PPO initializes above the cooperative optimum ($S{\approx}250$ vs $S{\approx}150$), and collapses plunge through the contested band too quickly to sample, biasing the conditioned drift toward survivors.

Non-survivor-biased re-probe (also negative). We tested the natural fix directly. We re-ran the identical sweep with the policy re-initialized at the cooperative optimum ($S{\approx}150$, actor mean biased to $a_0{=}-0.4$) so that both surviving and collapsing runs are sampled inside the contested band during early training, and added a second, initialization-robust *basin-flow* localizer (net early displacement of the policy mean from the optimum) alongside the restoring-drift statistic. The re-initialization de-biased the observable (the drift is now smooth and monotone in w rather than flat and noisy) and the collapse-fraction crossing again reproduced the boundary ($w_{\text{crit}} \approx 0.83$). But neither statistic *localizes* w_{crit}: the drift and basin-flow are monotone in w without a zero-crossing, and their steepest change falls near $w{\approx}0.6$, not at the boundary. The observable therefore tracks *where the policy sits* rather than *where the boundary is*, which the initialization change does not fix. We consequently report magnitude localization as genuinely open: the mechanism's sign-level and functional-form predictions stand, but a magnitude-level prediction of w_{crit} remains future work. The re-probe driver and analysis are released with the code.

7 Exploration-Noise Ablation

The mechanism predicts w_{crit} rises with exploration noise. Sweeping the PPO entropy coefficient at the structural condition (300k, 20 seeds, $w \in \{0.5, \ldots, 0.9\}$), the boundary rises monotonically: $w_{\text{crit}} = 0.685$ at ent=0 (reproducing the null), 0.690 at 0.001, 0.715 at 0.01, and beyond 0.05 the system collapses at every tested w (Table 7). **A binding-noise dissociation.** Repeating under MAPPO on the transfer substrates: on LBF the prediction replicates (w_{crit} rises $0.13 \to 0.37$ over ent $\in \{0, 0.05\}$, shift +0.24, non-overlapping endpoints); on MPE the boundary does *not* move (shift +0.05, overlapping CIs; pooled slope over 1,320 runs positive but not significant, $\beta{=}{+}0.47$, CI $[-3.4, +4.3]$). This is a dissociation predicted by the mechanism: exploration noise moves the boundary only when it is the binding term in the SNR. The supply chain and LBF have sparse, sharp landscapes where injected entropy dominates; MPE's dense distance-shaped reward carries high intrinsic exploration that swamps the added term. The effect appears exactly where the mechanism says it should and is absent where it says it should be, a sharper test than uniform replication.

Table 7: Entropy-coefficient ablation ($\alpha{=}\beta{=}1$, $\lambda{=}1$, 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs.

entropy coef.	w_{crit} (95% CI)
0.000	0.685 [0.656, 0.710]
0.001	0.690 [0.667, 0.715]
0.010	0.715 [0.689, 0.743]
0.050	> 1 (collapse at all w)

8 Threshold Robustness

We re-estimate w_{crit} on Exp A @300k under nine variants of the collapse definition, varying the $\bar{S}$ threshold over $\{8, 10, 12\}$ and the $\bar{\pi}$ threshold over $\{-102.5, -127.1, -152.5\}$. Three variants with $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1) and excluded. Across the remaining six estimable variants the maximum deviation from the baseline $w_{\text{crit}} = 0.666$ is 0.015, well below the CI width: the collapse definition is robust to $\pm 20\%$ threshold perturbation.

9 Notation and Abbreviations

Abbreviations. MARL (multi-agent reinforcement learning); CTDE (centralized training, decentralized execution); CPT (cumulative prospect theory); PPO (proximal policy optimization); MAPPO (multi-agent PPO with a centralized critic); MPE (multi-agent particle environments); LBF (Level-Based Foraging); SNR (signal-to-noise ratio); CI (confidence interval).

10 Claim–Evidence Ledger

Table 8 states how broadly each headline claim is supported. The loss-aversion direction is the generalizing core (three environments, two algorithms); the curvature sign and the downstream-minority cascade are supply-chain-specific, the latter framed as a falsifiable prediction for directed-flow domains.

Symbol	Meaning
w	reward centralization weight, $\in [0, 1]$
w_{crit}	critical w where $P(\text{collapse})=0.5$
λ	loss-aversion coefficient ($\lambda>1$)
α, β	CPT value-function curvature (gain/loss)
ρ	loss-domain share of marginal incentive, $\in [0, 1]$
g, c	marginal global benefit / local cost of cooperation
$D(w), \Theta(w)$	Langevin drift / diffusion of the policy mean
$\text{SNR}(w)$	$D(w)/\sqrt{\Theta(w)}$, escape signal-to-noise ratio
σ, η, B	exploration scale / learning rate / batch size
$\Delta U, \tau$	escape barrier / Kramers first-passage time
T	training horizon (environment steps)
$\bar{S}, \bar{\pi}$	mean base-stock / mean profit (collapse label)

Table 8: Each headline claim and the breadth of evidence behind it. Environments: SC (supply chain), MPE, LBF. Algorithms: PPO, MAPPO.

Claim	Envs	Algos
Sharp collapse boundary w_{crit} exists	SC, MPE, LBF	PPO, MAPPO
Metastability (w_{crit} slides with T)	SC, MPE, LBF	PPO, MAPPO
Loss aversion *raises* w_{crit} (dir.)	SC, MPE, LBF	PPO, MAPPO
Affine λ-law (form)	SC	PPO
Mechanism fixes sign a priori	SC, entropy abl.	PPO
Curvature *lowers* w_{crit} (sign)	SC (opp. on LBF)	PPO
Downstream-minority cascade	SC only	PPO

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